

UMESH CHANDRA TIRUMANI

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Summary

Pre-final year CS student with two production-deployed systems and startup internship experience. Built **codeslayer.me** (online judge, 50+ active users) independently and a **6-agent AI supply chain pipeline** at **Google Agentathon** (15,000+ participants). Selected **top 1.4% of 40,000+** for AlgoUniversity's engineering accelerator.

Education

Anurag University, Hyderabad

B.Tech in Computer Science & Engineering

2023 – Present

CGPA: 8.49/10

Work Experience

AlgoUniversity (Y-Combinator-backed)

May 2025 – July 2025

Software Engineer Co-op Intern

Remote

- Solo-engineered and deployed a production online judge serving **150+ students**: sandboxed C++/Python execution engine with hard **TLE/MLE/CE** detection, eliminating manual code review for instructors.
- Delivered **3 major features** (problem management, submission history, Rate Limiting) across Agile sprints, consistently hitting sprint deadlines at production quality.
- Owned end-to-end deployment across two independently deployable microservices; shipped the full platform within a **10-week** co-op timeline.

Projects

Code Slayer: Full-Stack Online Judge | *React.js, Node.js, Express.js, MongoDB, Docker, AWS* [Source Code](#) — [Live Demo](#)

- Deployed to production at **codeslayer.me** serving **50+ registered users** with **1,000+ submissions in the first month** across two independently deployable microservices on **Render** and **AWS EC2/ECR**.
- Engineered a sandboxed **multi-language execution engine** (C++, Python) with hard per-submission limits (**1,000ms timeout, 256MB RAM, 50MB output cap**), **JWT + bcrypt** auth, and indexed **MongoDB** schemas at **sub-200ms** query response.
- Built a **3-scenario Gemini AI reviewer** with custom prompts per verdict (Accepted, Wrong Answer, Edge Case Fail) covering bugs, complexity, and alternative approaches.

MarketPulse-X: AI Supply Chain Intelligence | *Python, FastAPI, LangGraph, Kubernetes (GKE)* [Source Code](#) — [Live Demo](#)

- Co-built at **Google Agentathon** (15,000+ participants, jury recognized); owned **LangGraph architecture, SEC EDGAR integration**, and **GKE deployment** of a 6-agent supply chain risk monitoring pipeline.
- Designed an **autonomous confidence loop** that detects low-confidence analyses (<70%), refines search queries, and re-triggers upstream agents up to 3 iterations; validated against real Yahoo Finance and NewsAPI data.
- Engineered a **SEC EDGAR 10-K parser** extracting supplier and customer relationships for any company without pre-configuration, fused with live news signals via **parallel LLM inference**.
- Deployed on **Docker + GKE** with horizontal pod autoscaling, delivering portfolio risk alerts via WebSocket at **sub-100ms** latency in production.

Technical Skills

Languages: Python, C++, JavaScript

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI

AI & LLM: Multi-agent pipelines, LangGraph

Databases & Cloud: MongoDB, MySQL, AWS (EC2, ECR), Google Cloud Platform (GKE), Render

DevOps & Tools: Docker, Kubernetes, WebSockets, REST APIs, Git, Postman

Achievements & Certifications

- **AlgoUniversity Accelerator** (Y-Combinator-backed) – Selected **top 1.4% of 40,000+** applicants; completed production-focused DSA camps with mentorship from senior industry engineers.
- **Google Cloud Skills Boost** – 25 skill badges spanning Gemini Application Development, Multimodal RAG, MLOps for Generative AI, and Responsible AI. [Public Profile](#)
- **Competitive Programming** – LeetCode: 250+ problems solved, rating **1497**. [LeetCode](#) — [HackerRank](#)
- **Hackathons** – Finalist/builder at **Google Agentathon** (15,000+ participants) and 3+ inter-college hackathons; shipped working prototypes under 24-hour deadlines.